

Lesson 4.1: Introducing Likelihoods

Decisions with likelihoods

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Uncertainty vs likelihoods

Example (River deliveries)

Alice has to deliver one package to C every day for some *future period*. Records show that the ferry was operating on 75 of the last 100 days.

'One shot' decision

Future period = one day (fuel consumed):

	$\bar{f}$	f	E
A	4	0	1.0
B	3	1	1.5

How to choose?

Fuel *saved* (4L tank):

	$\frac{1}{4}$ $\overline{f}$	$\frac{3}{4}$ $\overline{f}$	E	min	maxL
A	0	4	3.0	0	4
B	1	3	2.5	1	3

Short to long term

	$\frac{1}{16}$ $\overline{ff}$	$\frac{3}{16}$ $\overline{ff}$	$\frac{3}{16}$ $\overline{ff}$	$\frac{9}{16}$ $\overline{ff}$
A	0	2	2	4
B	1	2	2	3

	$\frac{1}{16}$ $\overline{ff}$	$\frac{6}{16}$ $\frac{1}{2}\overline{f}\frac{1}{2}\overline{f}$	$\frac{9}{16}$ $\overline{ff}$
A	0	2	4
B	1	2	3

	$\frac{1}{4}\overline{f}\frac{3}{4}\overline{f}$
A	3.0
B	2.5